θ = parameter

θ_0 = hypothesized value

Stat 266 – Unit 10 – Video Set 1 – Class Examples

Summary: Hypothesis Testing

Goal: To compare/test two competing claims (hypotheses) about unknown population parameters.

- ***Null*** Hypothesis ($H_0 \rightarrow$ "H naught"): This is usually a statement that our parameter(s) are equal to particular value(s), indicating no effect, no difference, no change, no consequence... $H_0: \theta = \theta_0$
- ***Alternative*** Hypothesis ($H_a \rightarrow$ "H - A"): This is often a statement that our parameter(s) are in a range of values, indicating some effect, some difference, some change, some consequence... e.g. $H_a: \theta > \theta_0$, $H_a: \theta \neq \theta_0$

research

Logic of the Test :

- Assume the null hypothesis (H_0) is true.
- Collect evidence (data). Based on the data, measure the strength of the evidence against H_0 in favour of H_a.

Decisions / Conclusions :

1. Reject H_0 in favour of H_a. You found enough evidence against H_0 in favour of H_a.
2. Fail to reject H_0 in favour of H_a. You did not find enough evidence against H_0 in favour of H_a.

Errors :

significance level of test.

1. Type I Error: Rejecting a true H_0. $\Rightarrow \alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$.
2. Type II Error: Failing to reject a false H_0. $\Rightarrow \beta = P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$.

$1 - \beta$
$= P(\text{reject } H_0 \mid H_0 \text{ is false})$
$=$ power of the test.

H_0: innocent
H_a: guilty

NEVER "accept" H_0.

Rejections Region (RR) Approach :

statistics

For some function of the sample, $U = g(Y_1, \ldots, Y_n)$, choose to reject H_0 in favour of H_a if $U \in RR$.

- $\alpha = P(U \in RR \mid H_0 \text{ is true})$. $= P(\text{reject } H_0 \mid H_0 \text{ is true})$
- $\beta = P(U \notin RR \mid H_0 \text{ is false})$. $= P(\text{fail to reject } H_0 \mid H_0 \text{ is false})$

Examples : $RR = \{U = \bar{Y} : \bar{y} < 15\}$, $RR = \left\{U = \prod_{i=1}^{n} Y_i : \prod_{i=1}^{n} y_i > k\right\}$, $RR = \{U = Y_{(k)} : 1 < y_{(k)} < 2\}$

The P - value :

$\Rightarrow$ ***p - value*** $= P($that another sample would produce just as much or more evidence against H_0 in favour of H_A than what was actually observed, if H_0 is true$)$.

- The ***p - value*** is used as a measure for the strength of the evidence against H_0 in favour of H_A.
- The smaller this probability, the stronger the evidence against H_0 in favour of H_a.

Statistical Significance :

$0 < \alpha < 1$: Common choices are $\alpha = 0.01, 0.05, 0.10$.

1. Choose a significance level α.
2. Sample and calculate the p - $value$.
3. Then draw your conclusion as follows:

- If p - $value \le \alpha$ then we have significant evidence against H_0 in favour of H_a at significance level α.
  With a significance level of α, we can reject H_0 in favour of H_a. $\Rightarrow$ does not imply H_0 is false
- If p - $value > \alpha$ then we have insignificant evidence against H_0 in favour of H_a at significance level α.
  With a significance level of α, we fail to reject H_0. $\Rightarrow$ does not imply H_0 is true

Example A:

Suppose $Y_1,\ldots,Y_n$ is a random sample of lifetimes of lightbulbs (in hundreds of hours). Suppose also that all bulbs are independent with lifetime described by the following probability density:

$$\Rightarrow f_Y(y)=\begin{cases}\frac{1}{\theta}e^{-y/\theta} & , y\geq 0\\ 0 & , \text{else}\end{cases}$$

$Y \sim Exp(\mu_Y = \theta)$

The company that manufactures the bulbs advertise that average lifetime is 300 hundred hours. A consumer group believes otherwise that average lifetime is under 300 hours.

The test set-up is: $H_0:\theta=3 \quad vs. \quad H_a:\theta<3$ $\Leftrightarrow$ $H_0: \underbrace{\mu_Y = 3}_{\text{or } \mu_Y \geq 3}$ vs. $H_a: \mu_Y < 3$

Based on n observations the consumer group has decided to reject the null hypothesis based on the following rejection region (rule): $RR=\{\bar{Y}:\bar{y}<k\}$.

a) Define α. Explore α for different values of n and k.

b) For fixed (chosen α), define k.

c) Define how the p-value is calculated.

a) $H_0: \theta = \theta_0$ vs $H_a: \theta < \theta_0$

$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true}) = P(\bar{Y} < k \mid \theta = \theta_0)$

- $Y \sim Exp(\theta)$, $\sum^n Y \sim Gamma(\alpha = n, \beta = \theta)$,
- $\frac{1}{n}\sum^n Y = \bar{Y} \sim Gamma(\alpha = n, \beta = \theta/n)$
- $\frac{2n}{\theta}\bar{Y} \sim Gamma(\alpha = n, \beta = 2) \sim \chi^2_{2n}$.

$$\alpha = P(\bar{Y} < k \mid \theta = \theta_0) = P\left(\frac{2n\bar{Y}}{\theta} < \frac{2nk}{\theta} \,\middle|\, \theta = \theta_0\right)$$

$$\boxed{\alpha = P\left(\chi^2_{2n} < \frac{2nk}{\theta_0}\right)}$$

e.g. Suppose $\theta_0 = 3$, and we choose $k = 2.50$ based on $n = 10$.

$$\alpha = P\left(\chi^2_{20} < \frac{2(10)(2.50)}{3}\right) = P(\chi^2_{20} < 16.667)$$

$$= 1 - \underbrace{P(\chi^2_{20} > 16.667)}_{\in (0.10,\, 0.90)} \in (0.10, 0.90)$$

Actual: 0.325

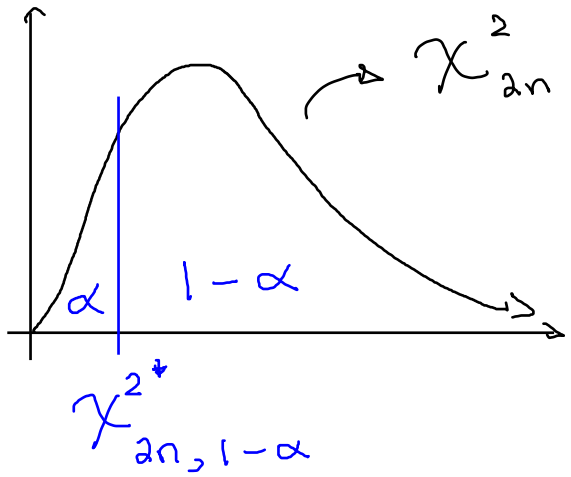
e.g. Suppose $\theta_0 = 3$, and we choose $k = 2.20$ based on $n = 50$.

$$\alpha = P\left(\chi^2_{100} < \frac{2(50)(2.20)}{3}\right) = P(\chi^2_{100} < 73.33)$$

$$= 1 - \underbrace{P(\chi^2_{100} > 73.33)}_{\in (0.975,\, 0.99)} \in (0.01, 0.025)$$

Actual: 0.0208

b) $\alpha = P\left(\chi^2_{2n} < \frac{2nk}{\theta_0}\right)$ $\Rightarrow$ For fixed α, what is k?



$$\Rightarrow P\left(\chi^2_{2n} < \chi^{2*}_{2n,1-\alpha}\right) = \alpha$$

$$\Rightarrow \frac{2nk}{\theta_0} = \chi^{2*}_{2n,1-\alpha} \Rightarrow \boxed{k = \frac{\theta_0}{2n}\chi^{2*}_{2n,1-\alpha}}$$

$$RR_{\alpha,n} = \left\{ \bar{Y} : \bar{y} < \frac{\theta_0}{2n}\chi^{2*}_{2n,1-\alpha} \right\}$$

e.g. Suppose $\theta_0 = 3$ and based on $n = 10$ we want $\alpha = 0.05$:

$\Rightarrow \chi^{2*}_{20,0.95} = 10.851$, $k = \frac{3}{20}(10.851) = 1.628$

$RR_{\alpha=0.05,\, n=10} = \{ \bar{Y} : \bar{y} < 1.628 \}$.

$\Rightarrow$ At $\alpha = 0.05$, based on $n = 10$, we can only conclude $\mu_Y < 3$ if $\bar{y} < 1.628$.

e.g. Suppose $\theta_0 = 3$ and we want $\alpha = 0.05$ based on $n = 50$:

$$k = \frac{\theta_0}{2n} \chi^{2*}_{2n,1-\alpha} = \frac{3}{2(50)} \chi^{2*}_{100,0.95}$$

$$= \frac{3}{100}(77.929) = 2.338$$

$$RR_{\alpha=0.05, n=50} = \{ \bar{y} : \bar{y} < 2.338 \}$$

c) Define the p-value: ($H_a: \theta < \theta_0$)

$$\text{p-value} = P(\bar{Y} < \bar{y} \mid \theta = \theta_0) = P\left(\frac{2n\bar{Y}}{\theta} < \frac{2n\bar{y}}{\theta} \mid \theta = \theta_0 \right)$$

$$= P\left(\chi^2_{2n} < \frac{2n\bar{y}}{\theta_0} \right).$$

$$\boxed{\text{p-value} = P\left(\chi^2_{2n} < \frac{2n\bar{y}}{\theta_0}\right)}$$

$$\boxed{RR_{\alpha,n} = \left\{ \bar{y} : \bar{y} < \frac{\theta_0}{2n} \chi^{2*}_{2n,1-\alpha} \right\}}$$

Suppose we want to carry out the test at $\alpha = 0.10$. $\theta_0 = 3$.

e.g. 1) Suppose we observed $\bar{y} = 2.90$ from $n = 10$.

- $\text{p-value} = P\left(\chi^2_{20} < \frac{2(10)(2.90)}{3}\right) = P(\chi^2_{20} < 19.333)$

$$= 1 - \underbrace{P(\chi^2_{20} > 19.333)}_{\in (0.10, 0.90)} \in (0.10, 0.90)$$

Actual ≈ 0.4997

- At $\alpha = 0.10$ (p-value $> \alpha$), we fail to reject H_0.

$\Rightarrow$ We have insignificant evidence to suggest that $\theta = \mu_Y < 3$.

- Note: $RR_{\alpha=0.10, n=10} = \left\{ \bar{Y} : \bar{y} < \frac{3}{20}(12.443) = 1.866 \right\}$

$\Rightarrow$ $\bar{y} = 2.90$ is NOT in the RR.

2) Suppose we observed $\bar{y} = 2.00$ from $n = 50$.

- p-value $= P\left(\chi^2_{100} < \frac{2(50)(2.00)}{3}\right) = P(\chi^2_{100} < 66.667)$

$$= 1 - \underbrace{P(\chi^2_{100} > 66.667)}_{> 0.995} < 0.005.$$

Actual: 0.00418.

- At $\alpha = 0.10$ (p-value $< \alpha$), we can reject H_0.

$\Rightarrow$ We have significant evidence to suggest that $\theta = \mu_Y < 3$.

Note: $RR_{\alpha=0.10, n=50} = \left\{ \bar{y} : \bar{y} < \frac{3}{100}(82.358) = 2.471 \right\}$.

$\Rightarrow$ $\bar{y} = 2.00$ IS in the rejection region.

Example B : Suppose you select a sample of n observations from a population with probability density

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{3} e^{-\frac{(y-\theta)}{3}} & , \quad \theta < y \\ 0 \ , & \text{else} \end{cases}$$

Parameter: θ = population minimum

$Y_{(1)}$ is minimally sufficient for θ.

You wish to carry out the following test based on the statistic $Y_{(1)}$.

$$H_0 : \theta = 2 \ vs. \ H_a : \theta > 2$$

The rejection region is $RR = \{Y_{(1)} > k\}$.

a) If you want to carry out the test based on $n = 10$ observations with significance level $\alpha = 0.01$, what is k?

b) Suppose from a sample of $n = 30$, you observed $y_{(1)}$ = 2.4. What is the p-value of the test?

$$F_Y(y) = P(Y \le y) = \int_\theta^y f_Y(t)\,dt = 1 - e^{-(y-\theta)/3} \ , \quad y \ge \theta .$$

$$\therefore \ f_{Y_{(1)}}(y) = n[1 - F_Y(y)]^{n-1} f_Y(y) = \frac{n}{3} e^{-\frac{n}{3}(y-\theta)} \ , \quad y > \theta$$

a) $\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true}) = P(Y_{(1)} > k \mid \theta = \theta_0)$

$$= \int_k^\infty \frac{n}{3} e^{-\frac{n}{3}(y-\theta_0)}\,dy = e^{-\frac{n}{3}(k-\theta_0)} .$$

For fixed α and n, what is k?

$$\Rightarrow k = \theta_0 - \frac{3}{n}\ln(\alpha). \quad \Rightarrow RR_{\alpha,n} = \left\{ Y_{(1)} : y_{(1)} > \theta_0 - \frac{3}{n}\ln(\alpha) \right\} .$$

For $\alpha = 0.01$, from $n = 10$ and $\theta_0 = 2$,

$$k = \theta_0 - \frac{3}{n}\ln(\alpha) = 2 - \frac{3}{10}\ln(0.01) \approx 3.3816.$$

$$RR_{\alpha=0.01,\, n=10} = \underbrace{\{ Y_{(1)} : y_{(1)} > 3.3816 \}}$$

We will only conclude θ (population minimum) is larger than 2 if $y_{(1)}$ (sample minimum) is larger than 3.3816.

b) $$p\text{-value} = P(\underbrace{Y_{(1)} > y_{(1)}}_{\text{more evidence against } H_0 \text{ in favour of } H_a:\, \theta > \theta_0} \mid \theta = \theta_0) = \int_{y_{(1)}}^{\infty} \frac{n}{3} e^{-\frac{n}{3}(y-\theta_0)}\, dy$$

$$= e^{-\frac{n}{3}(y_{(1)} - \theta_0)}.$$

If $\theta_0 = 2$ and we observed $y_{(1)} = 2.4$ from $n = 30$,

$$p\text{-value} = e^{-\frac{30}{3}(2.4-2)} = e^{-4} \approx 0.01832.$$

- At $\alpha = 0.01$ (p-value $> \alpha$), we fail to reject H_0 (we have insignificant evidence against H_0).

$$k = \theta_0 - \frac{3}{n}\ln(\alpha) = 2 - \frac{3}{30}\ln(0.01) \approx 2.4605$$

$$RR_{\alpha=0.01,\, n=30} = \underbrace{\{ Y_{(1)} : y_{(1)} > 2.4605 \}}_{y_{(1)} = 2.4 \text{ is NOT in the RR.}}$$

- At $\alpha = 0.05$ (p-value $< \alpha$), we can reject H_0 (we have significant evidence against H_0).

$$k = 2 - \frac{3}{30}\ln(0.05) = 2.2996$$

$$RR_{\alpha=0.05,\, n=30} = \underbrace{\{ Y_{(1)} : y_{(1)} > 2.2996 \}}_{y_{(1)} = 2.4 \text{ IS in the RR.}}$$

Example C : Suppose $Y_1, Y_2, \ldots, Y_n$ are iid r.v.'s with p.d.f

$$f_Y(y) = \begin{cases} \dfrac{3\theta^3}{y^4} & , 0 < \theta < y \\ 0 & , else \end{cases}$$

Parameter: θ = population minimum

$Y_{(1)}$ is minimally sufficient.

You wish to carry out the following test based on the statistic $Y_{(1)}$:

$$H_0 : \theta = \theta_0 \ vs. \ H_a : \theta > \theta_0.$$

The rejection region is $RR = \{Y_{(1)} : y_{(1)} > k\}$.

$$f_{Y_{(1)}}(y) = \frac{3n\theta^{3n}}{y^{3n+1}}, \quad 0 < \theta < y$$

a) Suppose $n = 15$. If $k = 1.10\theta_0$, what is α?

b) Suppose $n = 20$. If you found the p-value to be 0.095, what was the observed value of $y_{(n)}$.

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ is true}) = P(Y_{(1)} > k \mid \theta = \theta_0)$$

$$= \int_k^\infty \frac{3n\theta_0^{3n}}{y^{3n+1}}\, dy \quad \text{OR} \quad 1 - F_{Y_{(1)}}(k)$$

$$= \left(\frac{\theta_0}{k}\right)^{3n}.$$

$$\Rightarrow \alpha = \left(\frac{\theta_0}{1.10\theta_0}\right)^{3(15)} = \left(\frac{1}{1.10}\right)^{45} = 0.01372.$$

For fixed α, $k = \theta_0(\alpha)^{-1/3n}$.

$$RR_{\alpha,n} = \{Y_{(1)} : y_{(1)} > \theta_0(\alpha)^{-1/3n}\}.$$

e.g. If $\alpha = 0.05$, $n = 20$, and $\theta_0 = 5$,

$$RR_{\substack{\alpha=0.05 \\ n=20}} = \{Y_{(1)} : y_{(1)} > 5.256\}.$$

b) Suppose $H_0: \theta = 5$ vs $H_a: \theta > 5$.

Suppose $n = 20$ and we ended up with a p-value $= 0.095$, what was $y_{(1)}$?

$$\text{p-value} = P(Y_{(1)} > y_{(1)} \mid \theta = \theta_0)$$

$$= \int_{y_{(1)}}^{\infty} \frac{3n\,\theta_0^{3n}}{y^{3n+1}}\,dy = \left(\frac{\theta_0}{y_{(1)}}\right)^{3n}$$

Note: If $\theta_0 > y_{(1)}$ there is NO evidence against H_0. $\Rightarrow$ p-value $= 1$.

$$= \left(\frac{5}{y_{(1)}}\right)^{60} \Rightarrow 0.095$$

$$\Rightarrow y_{(1)} = \frac{5}{0.095^{1/60}} \approx 5.200.$$

Note: p-value $= 0.095 > \alpha = 0.05$ (fail to reject H_0).

$$RR_{\alpha=0.05,\, n=20} = \{\underbrace{Y_{(1)} : y_{(1)} > 5.256}\}$$

$y_{(1)} = 5.200$ is NOT in the RR.

Note: If $\alpha = 0.10$, (p-value $< \alpha$), we can reject H_0.

$$RR_{\alpha=0.10,\, n=20} = \{Y_{(1)} : y_{(1)} > 5(0.10)^{-1/60} = 5.196\}$$

($y_{(1)} = 5.200$ is in the RR).

Example D : Suppose $\{Y_1,\ldots,Y_n\}$ is a random sample of independent and identically distributed $Uniform(0,\theta)$ random variables.

Parameter: θ = population maximum

$$\Rightarrow f_Y(y)=\begin{cases}\dfrac{1}{\theta} & , 0<y<\theta \\ 0 & , else\end{cases}$$

$Y_{(n)}$ is minimally sufficient.

You wish to carry out the following test based on the statistic $Y_{(n)}$.

$$H_0:\theta=2 \; vs. \; H_a:\theta>2$$

$$f_{Y_{(n)}}(y) = \frac{n y^{n-1}}{\theta^n}, \quad 0<y<\theta$$

The rejection region is $RR=\{Y_{(n)}: y_{(n)}>k\}$.

a) If you want to carry out the test based on $n=15$ observations with significance level $\alpha=0.01$, what is k?

b) Suppose from a sample of $n=30$, you observed $y_{(n)}=1.90$. What is the p-value of the test?